



The Spotify Data Analysis SQL Project Using PostgreSQL

The **Spotify Data Analysis SQL Project** is a comprehensive data analytics portfolio project designed to explore, clean, and extract meaningful business insights from a Spotify dataset. The end-to-end journey encompassed downloading raw data from Kaggle, performing data inspection and cleaning (such as handling anomalies like zero-duration tracks), and importing the structured dataset into a PostgreSQL database.

To address real-world analytical scenarios, the project formulated and solved 15 structured business questions divided into three distinct difficulty tiers (Easy, Intermediate, and Advanced) using advanced SQL queries.

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-- Q1 : Retrieve the names of all tracks that have more than 1 billion streams.

SELECT track, stream

FROM spotify

WHERE stream > 1000000000;

Data Output

Messages

Explain X

Notifications

Showing rows: 1 to 385

Page No: 1 of 1

	track character varying (255)	stream bigint
1	Feel Good Inc.	1040234854
2	Californication	1055738398
3	Under the Bridge	1061750522
4	In Da Club	1041736808
5	Yellow	1483622308
6	Viva La Vida	1325575712
7	Something Just Like This	2030825694
8	The Scientist	1465980600
9	A Sky Full of Stars	1021782403
10	Hymn for the Weekend	1049654852
11	Fix You	1115171056
12	In the End	1444841705
13	Numb	1195213410
14	Creep	1101854725
15	Highway to Hell	1140164505

Total rows: 385

Query complete 00:00:00.136

CRLF

Ln 94, Col 61

```

94 -- Q2 : List all albums along with their respective artists.
95
96 SELECT DISTINCT album, artist albums
97 FROM spotify
98 ORDER BY album;

```

Data Output Messages Explain X Notifications

Showing rows: 1 to 1000 SQL

	album character varying (255)	albums character varying (255)
1	'Justments	Bill Withers
2	'N Sync UK Version	*NSYNC
3	'The Sounds of Night...	Nightwish
4	- TRAGEDY +	\$NOT
5	!Volare! The Very Best ...	Gipsy Kings
6	"Awaken, My Love!"	Childish Gambino
7	"Heroes" (2017 Remas...	David Bowie
8	"Let's Rock"	The Black Keys
9	"Let Go" Dj Pack	Dina Rae
10	"Miguel"	Miguel Bosé
11	#1 (Colby O'Donis Re...	Colby O'Donis
12	#1s ... and then some	Brooks & Dunn
13	#3	The Script
14	#FREESANTANA	Juelz Santana

Total rows: 14178 Query complete 00:00:00.263

CRLF

Ln 93, Col 1

Query

Query History

101

-- Q3 : Get the total number of comments for tracks where licensed = TRUE.

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103

SELECT SUM(comments) AS total_comments

104

FROM spotify

105

WHERE licensed = TRUE

106

GROUP BY licensed;

Data Output

Messages

Explain X

Notifications

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SQL

Showing rows: 1 to 1

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Page No: 1

of 1

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	total_comments numeric
1	497015695

```

110 -- Q4 : Find all tracks that belong to the album type single.
111
112 SELECT track, album_type FROM spotify
113 WHERE album_type = 'single';
114
115 |

```

Data Output Messages Explain X Notifications

Showing rows: 1 to 1000 Page No: 1 of 5

	track character varying (255)	album_type character varyi
1	New Gold (feat. Tame Im...	single
2	New Gold (feat. Tame Im...	single
3	Cracker Island (feat. Thun...	single
4	Best Friend - Remix	single
5	Lux Æterna	single
6	Screaming Suicide	single
7	Get Lucky (feat. Pharrell ...	single
8	Numb / Encore	single
9	Pump It Louder	single
10	Never Gonna Not Dance ...	single
11	Raise Your Glass	single
12	TRUSTFALL	single
13	Down In Atlanta	single
14	Cash In Cash Out	single
15	...	...

Total rows: 4973 Query complete 00:00:00.158

CRLF Ln 115, Col 1

```
117 -- Q5 : Count the total number of tracks by each artist.
118
119 SELECT artist, COUNT(track) AS total_tracks FROM spotify
120 GROUP BY artist;
121 |
```

Data Output Messages Explain × Notifications

Showing rows: 1 to 1000

Page No: 1

of 3

	artist character varying	total_tracks bigint
1	Lofi Fruits Mu...	10
2	Ray Charles	10
3	Joey Bada\$\$	10
4	La Mosca Tse...	10
5	TheFatRat	10
6	Leo Dan	10
7	Tyga	10
8	Grupo Marca ...	10
9	Tyler, The Cre...	10
10	NCT DREAM	10
11	Last Child	10
12	Plan B	10
13	Afgan	10
14	James Horner	10
15	James & Bond	10

Total rows: 2074 Query complete 00:00:00.372

CRLF

Ln 121, Col 1

```

135 |-- Q6 : Calculate the average danceability of tracks in each album.
136
137 SELECT album, AVG(danceability) AS avg_danceability
138 FROM spotify
139 GROUP BY album
140 ORDER BY avg_danceability DESC;

```

Data Output Messages Explain × Notifications

SQL

Showing rows: 1 to 1000

Page No: 1

of 12

	album character varying (255)	avg_danceability double precision
1	Funky Friday	0.975
2	FOR CERTAIN (Deluxe)	0.975
3	The House Is Burning ...	0.971
4	Quality Control: Contr...	0.97
5	N9NA	0.97
6	Aka Entre el Humo	0.967
7	Peru (Remix)	0.966
8	Imdat (feat. Hadise)	0.964
9	Street Gossip	0.964
10	Freddie's Inferno - Gho...	0.963
11	Tukoh Taka (Official FI...	0.963
12	Harder Than Ever	0.963
13	Losing Interest	0.96
14	Lucky Day	0.957
15	Memories	0.956

Total rows: 11853

Query complete 00:00:00.173

CRLF

Ln 135, Col 1

Query Query History

```
144  |-- Q7 : Find the top 5 tracks with the highest energy values.
145
146  SELECT track, MAX(energy) AS highest_energy
147  FROM spotify
148  GROUP BY track
149  ORDER BY highest_energy DESC
150  LIMIT 5;
```

Data Output Messages Explain X Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	track character varying (255)	highest_energy double precision
1	Rain and Thunderstorm, ...	1
2	Rain and Thunderstorm, ...	1
3	Rain and Thunderstorm, ...	1
4	Rain and Thunderstorm, ...	1
5	Gentle Piano Melodies	1

```

154 |-- Q8 : List all tracks along with their views and likes where official_video = TRUE.
155
156 SELECT track, SUM(views) AS total_views, SUM(likes) AS total_likes FROM spotify
157 WHERE official_video = TRUE
158 GROUP BY track
159 ORDER BY 2 DESC;
160

```

Data Output Messages Explain X Notifications

Showing rows: 1 to 1000 Page No: 1 of 14

	track character varying (255)	total_views double precision	total_likes numeric
1	Despacito	16159296273	101577278
2	See You Again (feat. Char...	11547595554	80295292
3	Shape of You	5908398479	31047780
4	Calma - Remix	5322011392	25649519
5	This Is What You Came For	5252059812	21207312
6	Swalla (feat. Nicki Minaj ...	5162403752	28864162
7	Wheels on the Bus	4898831101	14396841
8	Taki Taki (with Selena Go...	4828403435	37552053
9	Uptown Funk (feat. Bruno...	4821016218	20067879
10	Thrift Shop (feat. Wanz)	4508467245	30242165
11	Échame La Culpa	4491334603	20006575
12	Something Just Like This	4236043204	20565056
13	Sin Pijama	4172961620	17249780
14	Somebody That I Used To	4114890942	29168472

Total rows: 13650 Query complete 00:00:00.181

CRLF

Ln 154, Col 1

```
163 |-- Q9 : For each album, calculate the total views of all associated tracks.
```

```
164
```

```
165 SELECT album, track, SUM(views) AS total_views
```

```
166 FROM spotify
```

```
167 GROUP BY album, track
```

```
168 ORDER BY 3 DESC;
```

```
169
```

Data Output Messages Explain × Notifications

Showing rows: 1 to 1000

Page No: 1

of 19

	album character varying (255)	track character varying (255)	total_views double precision
1	VIDA	Despacito	16159296273
2	See You Again (feat. C...	See You Again (feat. Char...	11547595554
3	Peace Is The Mission ...	Lean On	9974504694
4	÷ (Deluxe)	Shape of You	5908398479
5	MUNAY	Calma - Remix	5322011392
6	This Is What You Cam...	This Is What You Came For	5252059812
7	Swalla (feat. Nicki Min...	Swalla (feat. Nicki Minaj ...	5162403752
8	CoComelon Kids Hits, ...	Wheels on the Bus	4898831101
9	Taki Taki (with Selena ...	Taki Taki (with Selena Go...	4828403435
10	Uptown Special	Uptown Funk (feat. Bruno...	4821016218
11	Gangnam Style (?????)	Gangnam Style (?????)	4679767471
12	The Heist	Thrift Shop (feat. Wanz)	4508467245
13	VIDA	Échame La Culpa	4491334603
14	Memories... Do Not On...	Something Just Like This	4236043204

Total rows: 18680

Query complete 00:00:00.175

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Ln 163, Col 1

```

171 |-- Q10 : Retrieve the track names that have been streamed on Spotify more than YouTube.
172 |SELECT * FROM
173 |    (SELECT track,
174 |         COALESCE(SUM(CASE WHEN most_played_on = 'Youtube' THEN stream END),0) AS most_played_on_youtube,
175 |         COALESCE(SUM(CASE WHEN most_played_on = 'Spotify' THEN stream END),0) AS most_played_on_spotify
176 |    FROM spotify
177 |   GROUP BY 1)
178 |WHERE (most_played_on_youtube < most_played_on_spotify)
179 |AND
180 |(most_played_on_youtube <> 0);

```

Data Output Messages Explain X Notifications

Showing rows: 1 to 155 Page No: 1 of 1

	track character varying (255)	most_played_on_youtube numeric	most_played_on_spotify numeric
1	Usted	30059201	137916795
2	3 A.M.	31174412	607872704
3	21 Hungarian Dances, Wo...	39575743	79151486
4	Mientes Tan Bien	6915455	224299945
5	Have You Ever Seen The ...	61903001	975300588
6	Dream A Little Dream Of ...	157256901	495674374
7	When I Grow Up	231236307	260959663
8	Me Hace Tanto Bien	56694580	187498268
9	What You Want (feat. Tot...	13099909	85458315
10	Suavemente	134326707	250550159
11	Emotions	218177	107654866

Total rows: 155 Query complete 00:00:00.323

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Ln 171, Col 1

```

197  |-- Q11 : Find the top 3 most-viewed tracks for each artist using window functions.
198
199  WITH ranking_artist AS (
200  SELECT
201      artist,
202      track,
203      SUM(views) AS total_views,
204      DENSE_RANK() OVER(PARTITION BY artist ORDER BY SUM(views) DESC) AS ranking
205  FROM spotify
206  GROUP BY 1,2
207  )
208  SELECT * FROM ranking_artist
209  WHERE ranking <= 3;

```

Data Output Messages Explain × Notifications

Showing rows: 1 to 1000 Page No: 1 of 7

	artist character varying	track character varying (255)	total_views double precision	ranking bigint
1	\$NOT	Tell Em	41100657	1
2	\$NOT	Like Me (feat. iann dior)	15803517	2
3	\$NOT	Mean	13563870	3
4	\$uicideboy\$	Paris	175156959	1
5	\$uicideboy\$	For the Last Time	91771038	2
6	\$uicideboy\$	Kill Yourself (Part III)	31674988	3
7	(G)I-DLE	POP/STARS	541864471	1
8	(G)I-DLE	TOMBOY	204797335	2
9	(G)I-DLE	Oh my god	191587399	3

Total rows: 6808

Query complete 00:00:00.259

CRLF

Ln 197, Col 1

```
213 |-- Q12 : Write a query to find tracks where the liveness score is above the average.
```

```
214
215 SELECT artist,
216        track,
217        liveness
218 FROM spotify
219 WHERE liveness > (SELECT AVG(liveness) FROM spotify);
220
```

Data Output Messages Explain X Notifications

Showing rows: 1 to 1000 SQL

	artist character varying	track character varying (255)	liveness double precis
1	Gorillaz	Feel Good Inc.	0.613
2	Gorillaz	DARE	0.298
3	Gorillaz	New Gold (feat. Tame Im...	0.325
4	Gorillaz	Cracker Island (feat. Thun...	0.325
5	Gorillaz	Dirty Harry	0.672
6	Red Hot Chili ...	Dani California	0.346
7	50 Cent	Candy Shop	0.38
8	50 Cent	Just A Lil Bit	0.315
9	50 Cent	Disco Inferno	0.279
10	Metallica	The Unforgiven (Remaste...	0.255
11	Metallica	One (Remastered)	0.42
12	Metallica	Screaming Suicide	0.239
13	Coldplay	Yellow	0.234

Total rows: 6364 Query complete 00:00:00.175

CRLF

Ln 213, Col 1

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223  -- Q13 : Use a WITH clause to calculate the difference between the highest and lowest energy values for tracks in each album
224
225  WITH energy_rank AS
226  (SELECT
227    album,
228    MAX(energy) AS highest_energy,
229    MIN(energy) AS lowest_energy
230  FROM spotify
231  GROUP BY 1
232  )
233  SELECT
234    album,
235    (highest_energy - lowest_energy) AS energy_diff
236  FROM energy_rank
237  ORDER BY energy_diff DESC;

```

Data Output Messages Explain × Notifications



Showing rows: 1 to 1000

Page No:

1

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	album character varying (255)	energy_diff double precision
1	White Noise	0.9067500000000001
2	Spotify Singles - Holid...	0.8360000000000001
3	Spotify Singles	0.8232
4	UNDERTALE Soundtra...	0.816
5	Making Mirrors	0.8109000000000001
6	Everytime We Touch (...)	0.8049999999999999
7	If I Can Dream: Elvis P...	0.787

Total rows: 11853

Query complete 00:00:00.144

CRLF

Ln 223, Col 1

```

241 |-- Q14 : Find tracks where the energy-to-liveness ratio is greater than 1.2.
242
243 SELECT
244     artist,
245     track,
246     energy_liveness
247 FROM spotify
248 WHERE energy_liveness > 1.2;
249

```

Data Output Messages Explain X Notifications

Showing rows: 1 to 1000 Page No: 1 of 19

	artist character varying	track character varying (255)	energy_liveness double precision
1	Gorillaz	Rhinestone Eyes	15.18358531
2	Gorillaz	New Gold (feat. Tame Im...	7.956896552
3	Gorillaz	On Melancholy Hill	11.546875
4	Gorillaz	Clint Eastwood	9.94269341
5	Gorillaz	DARE	2.989932886
6	Gorillaz	New Gold (feat. Tame Im...	2.76
7	Gorillaz	She's My Collar (feat. Kali...	7.276785714
8	Gorillaz	Cracker Island (feat. Thun...	2.809230769
9	Gorillaz	Dirty Harry	1.305059524
10	Red Hot Chili ...	Californication	6.039370079
11	Red Hot Chili ...	Under the Bridge	2.446808511
12	Red Hot Chili ...	Can't Stop	5.616766467

Total rows: 18797 Query complete 00:00:00.172

CRLF

Ln 241, Col 1

```
252 |-- Q15 : Calculate the cumulative sum of likes for tracks ordered by the number of views, using window functions.
```

```
253
254 SELECT
255     artist,
256     track,
257     album,
258     likes,
259     SUM(likes) OVER(ORDER BY views DESC) AS cumulative_likes,
260     views
261 FROM spotify;
```

Data Output Messages Explain X Notifications

Showing rows: 1 to 1000 Page No: 1 of 21

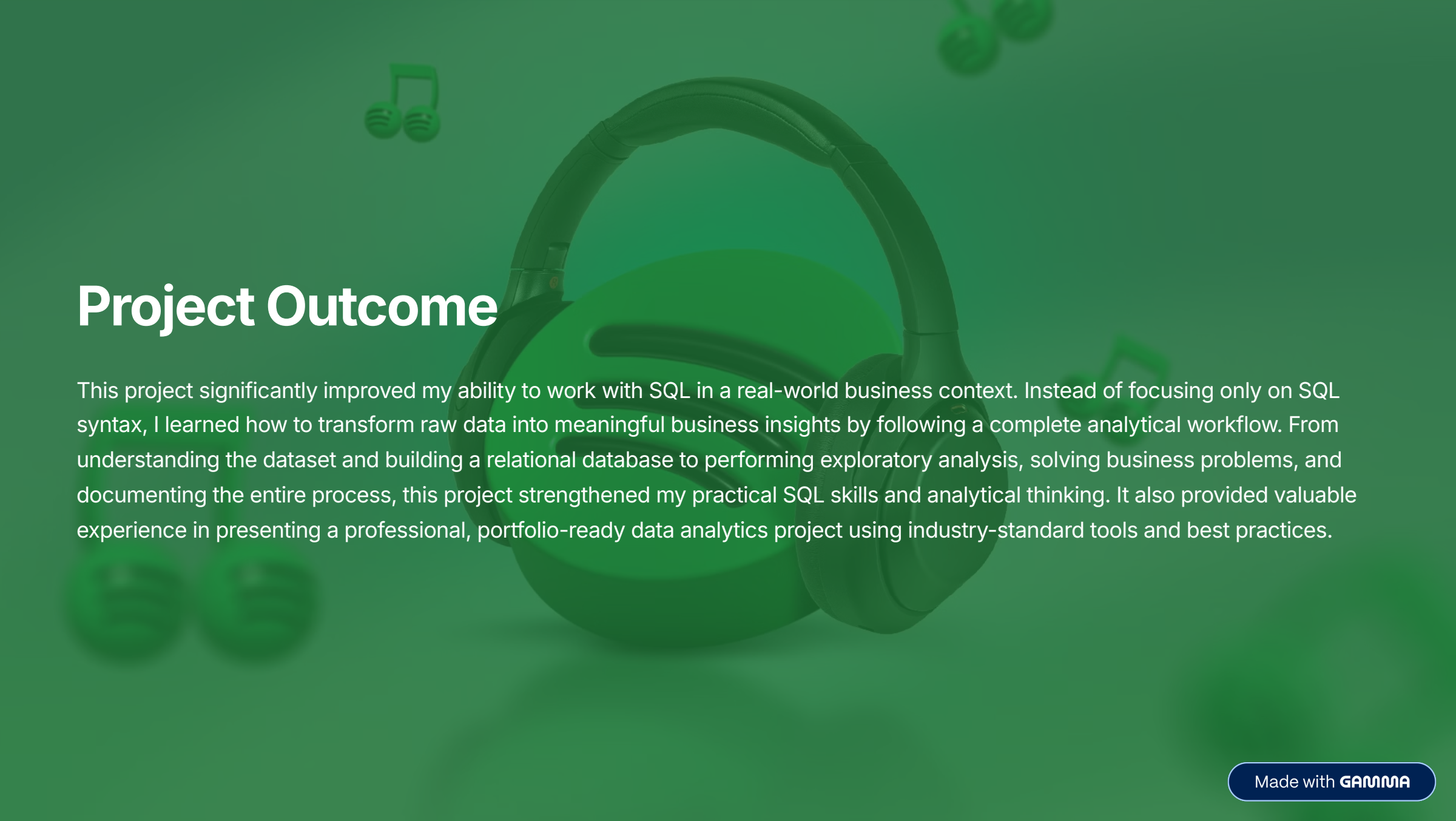
	artist character varying	track character varying (255)	album character varying (255)	likes bigint	cumulative_likes numeric	views double precision
1	Luis Fonsi	Despacito	VIDA	50788652	50788652	8079649362
2	Daddy Yankee	Despacito	VIDA	50788626	101577278	8079646911
3	Ed Sheeran	Shape of You	÷ (Deluxe)	31047780	132625058	5908398479
4	Charlie Puth	See You Again (feat. Char...	See You Again (feat. C...	40147674	172772732	5773798407
5	Wiz Khalifa	See You Again (feat. Char...	See You Again (feat. C...	40147618	212920350	5773797147
6	CoComelon	Wheels on the Bus	CoComelon Kids Hits, ...	14396841	227317191	4898831101
7	Mark Ronson	Uptown Funk (feat. Bruno...	Uptown Special	20067879	247385070	4821016218
8	PSY	Gangnam Style (?????)	Gangnam Style (?????)	26399133	273784203	4679767471
9	Maroon 5	Sugar	V	15340646	289124849	3817733132
10	Katy Perry	Roar	PRISM	15864499	304989348	3725748519
11	OneRepublic	Counting Stars	Native	16558621	321547969	3721610307

Total rows: 20592

Query complete 00:00:00.308

CRLF

Ln 252, Col 1



Project Outcome

This project significantly improved my ability to work with SQL in a real-world business context. Instead of focusing only on SQL syntax, I learned how to transform raw data into meaningful business insights by following a complete analytical workflow. From understanding the dataset and building a relational database to performing exploratory analysis, solving business problems, and documenting the entire process, this project strengthened my practical SQL skills and analytical thinking. It also provided valuable experience in presenting a professional, portfolio-ready data analytics project using industry-standard tools and best practices.